SOP: Monthly HIE Facility Data-Quality & Compliance Report

Synthetic data. All facilities, counts, and errors in this project are fictional and generated by `scripts/generate_data.py`. Nothing here reflects real NC HealthConnex participants, systems, or data.

1. Purpose

Produce a repeatable monthly view of (a) which participating facilities are sending data and (b) how clean that data is, so non-reporting and high-error facilities are caught and escalated early.

2. Source and cadence

Monthly extracts summarizing HL7 feed activity per facility: records received, last-received date, and validation errors by type. This project works at the extract level and does not parse raw HL7 messages. Run once per month after month-end, as soon as all three extracts are available.

3. Monthly run checklist

1. Place the new extracts in `data/raw/` (`facilities.csv`, `monthly_submissions_raw.csv`, `validation_errors_raw.csv`). Archive last month's files first.
2. Run `python scripts/run_pipeline.py`. It loads the raw files, runs `sql/00` to `sql/04`, and writes CSVs to `data/clean/`.
3. Open `validation_summary.csv` and review checks V01 to V10 (see section 4).
4. Open `rejected_rows.csv`. Every quarantined row needs a disposition: request a corrected extract, fix the facility master, or accept and document.
5. Reconcile: the sum of `total_records` in `monthly_summary.csv` should equal the clean record counts from the extract after quarantined rows are removed.
6. Refresh the Tableau workbook from `data/clean/`. Confirm the "Data as of" label shows the new month.
7. Export the dashboard to PDF, save it with the month in the file name, and log any escalations (section 6).

4. Validation rules and thresholds

ID	Rule	Handling
V01	Facility ID must match `FAC-###`	Standardized (case, spaces, hyphen)
V02	Month must be `YYYY-MM`	`MM/YYYY` and `Mon YYYY` converted
V03	Received date must be `YYYY-MM-DD`	`MM/DD/YYYY` converted
V04	Record count must not be blank	Row quarantined; facility-month marked Data Incomplete
V05	Record count between 0 and 500,000	Row quarantined; facility-month marked Data Incomplete
V06	One row per facility per month	Keep latest received date, then larger count
V07/V08	Facility ID must exist in the facility master	Row quarantined; follow up with feed owner
V09	Expected facility-months with no data or zero records	Flagged Silent
V10	Facilities with no data for more than 30 days	Flagged Silent 30+ days

Thresholds live in `sql/00_config.sql`: error rate above **3%** = Elevated Errors; no data for more than **30 days** = Silent 30+ days. Both are illustrative and should be set by program policy.

5. Metric definitions

- **% reporting:** facilities with at least one record that month ÷ facilities onboarded by month-end.
- **% compliant:** facilities reporting with an error rate at or below the threshold ÷ facilities onboarded by month-end.
- **Error rate:** validation errors ÷ records received, counting only facilities that reported.
- **Silent 30+ days:** days from the facility's last received date to month-end exceeds 30.
- **Worst offenders:** top 10 facilities by error rate over the trailing 3 months.

6. Escalation (sample process)

- **Silent 30+ days:** contact the facility's technical contact to confirm feed status and expected restart date. Still silent at the next monthly run: escalate to the program lead.
- **Elevated Errors two months in a row:** send the facility its error counts by type and ask for an EHR mapping review.
- **Unknown facility IDs or repeated duplicates:** raise with the feed owner or integration team.

This escalation path is a sample. Real contacts and timelines would come from program policy.

7. Limitations

Synthetic data; extract-level only; thresholds and escalation steps are illustrative; no PHI is used or needed.